NET : NEUROENDOCRINE TUMOURS

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NET Patient Foundation

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- Why look at NETs ?
- What is cancer - are NETs cancer ?
- What are NETs - where do they occur ?
- How are they classified ?
- Implications for prognosis
- Real life application
- Sources for further information

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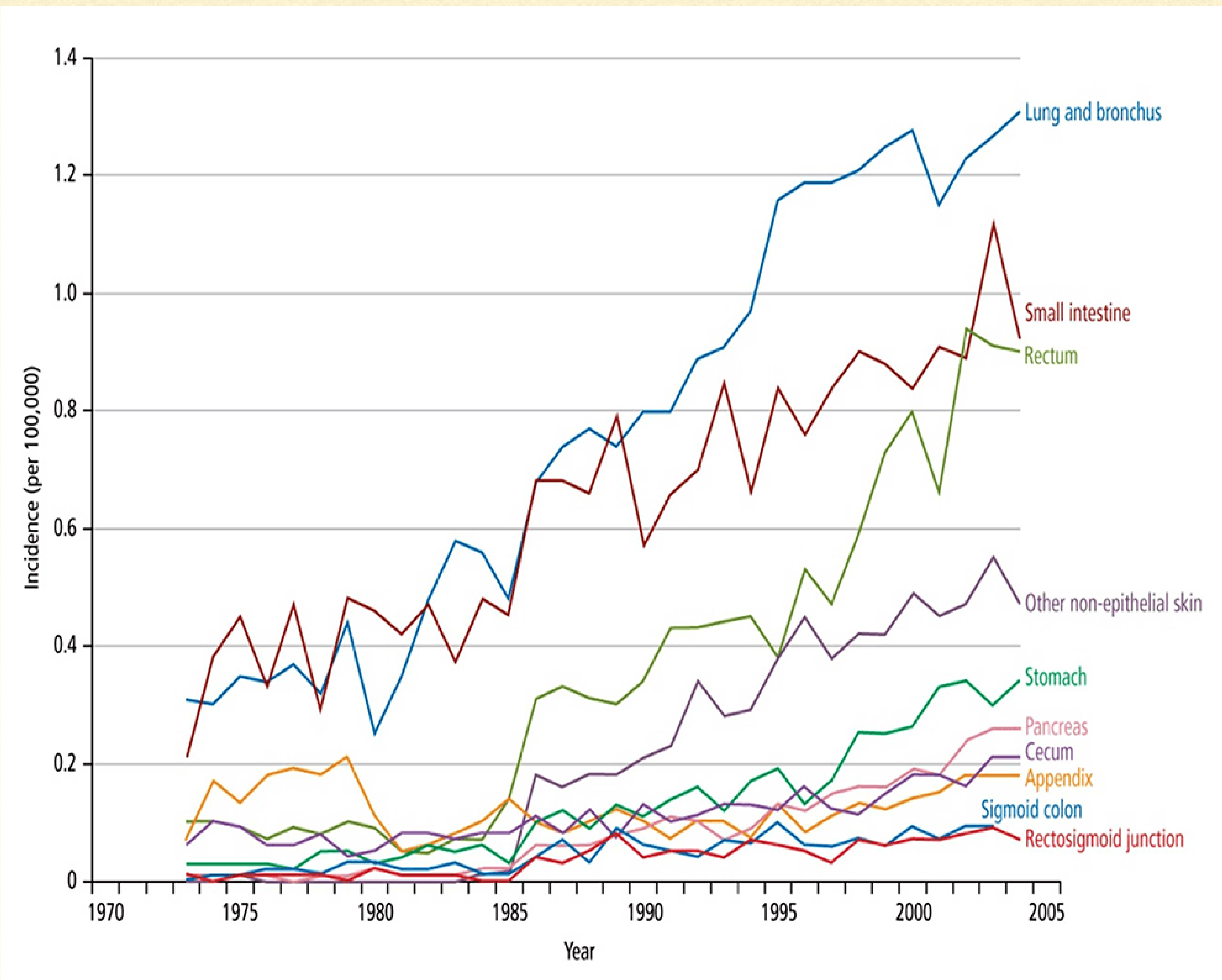
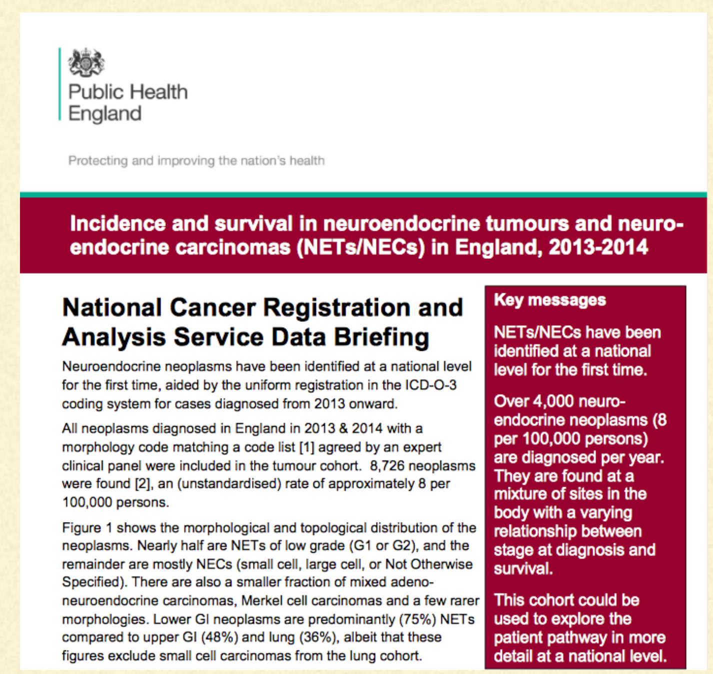


Global increase in incidence

nb high prevalence

In UK - PHE data



Public Health England

Protecting and improving the nation's health

Incidence and survival in neuroendocrine tumours and neuro-endocrine carcinomas (NETs/NECs) in England, 2013-2014

National Cancer Registration and Analysis Service Data Briefing

Neuroendocrine neoplasms have been identified at a national level for the first time, aided by the uniform registration in the ICD-O-3 coding system for cases diagnosed from 2013 onward.

All neoplasms diagnosed in England in 2013 & 2014 with a morphology code matching a code list [1] agreed by an expert clinical panel were included in the tumour cohort. 8,726 neoplasms were found [2], an (unstandardised) rate of approximately 8 per 100,000 persons.

Figure 1 shows the morphological and topological distribution of the neoplasms. Nearly half are NETs of low grade (G1 or G2), and the remainder are mostly NECs (small cell, large cell, or Not Otherwise Specified). There are also a smaller fraction of mixed adeno-neuroendocrine carcinomas, Merkel cell carcinomas and a few rarer morphologies. Lower GI neoplasms are predominantly (75%) NETs compared to upper GI (48%) and lung (36%), albeit that these figures exclude small cell carcinomas from the lung cohort.

Key messages

NETs/NECs have been identified at a national level for the first time.

Over 4,000 neuro-endocrine neoplasms (8 per 100,000 persons) are diagnosed per year. They are found at a mixture of sites in the body with a varying relationship between stage at diagnosis and survival.

This cohort could be used to explore the patient pathway in more detail at a national level.

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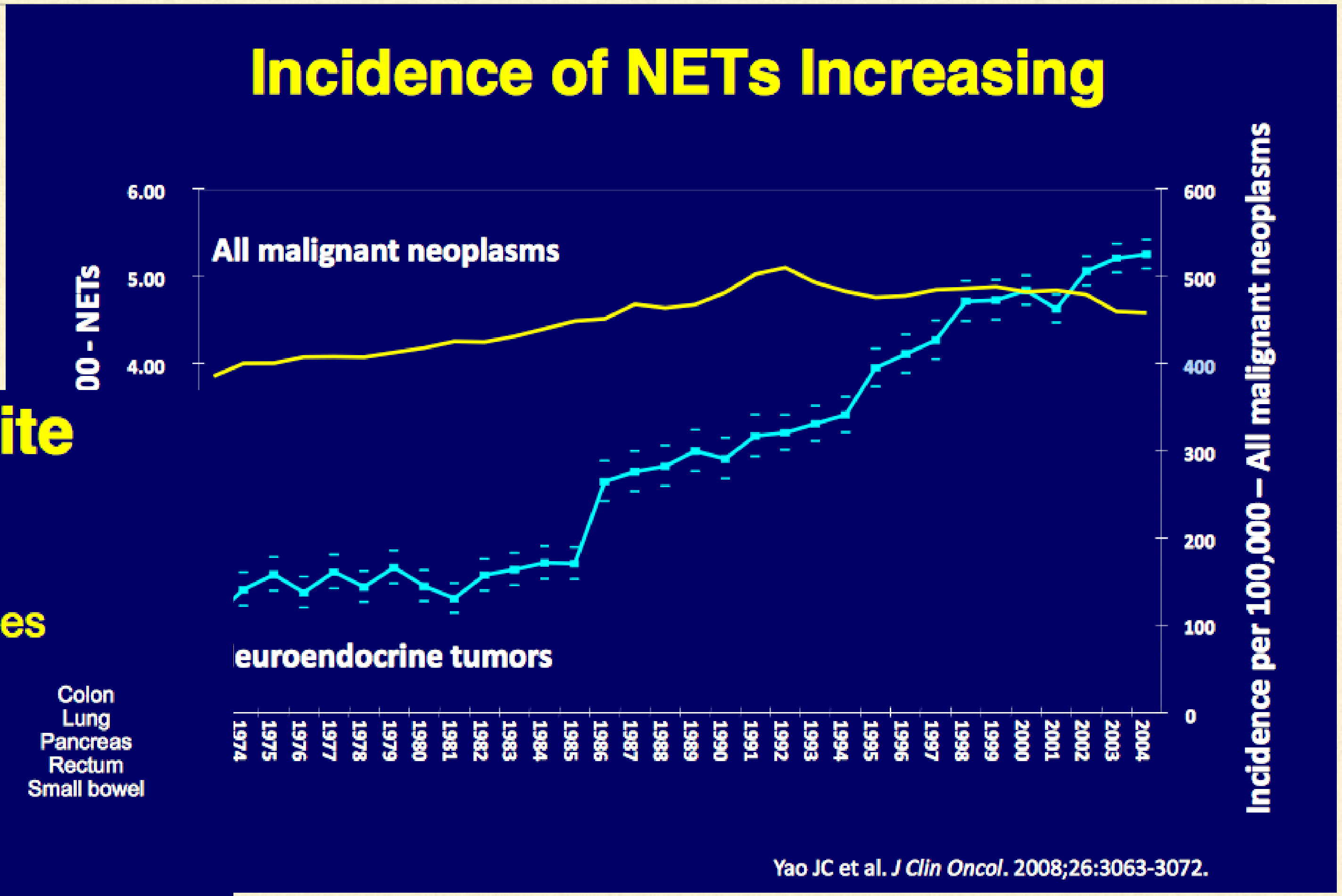

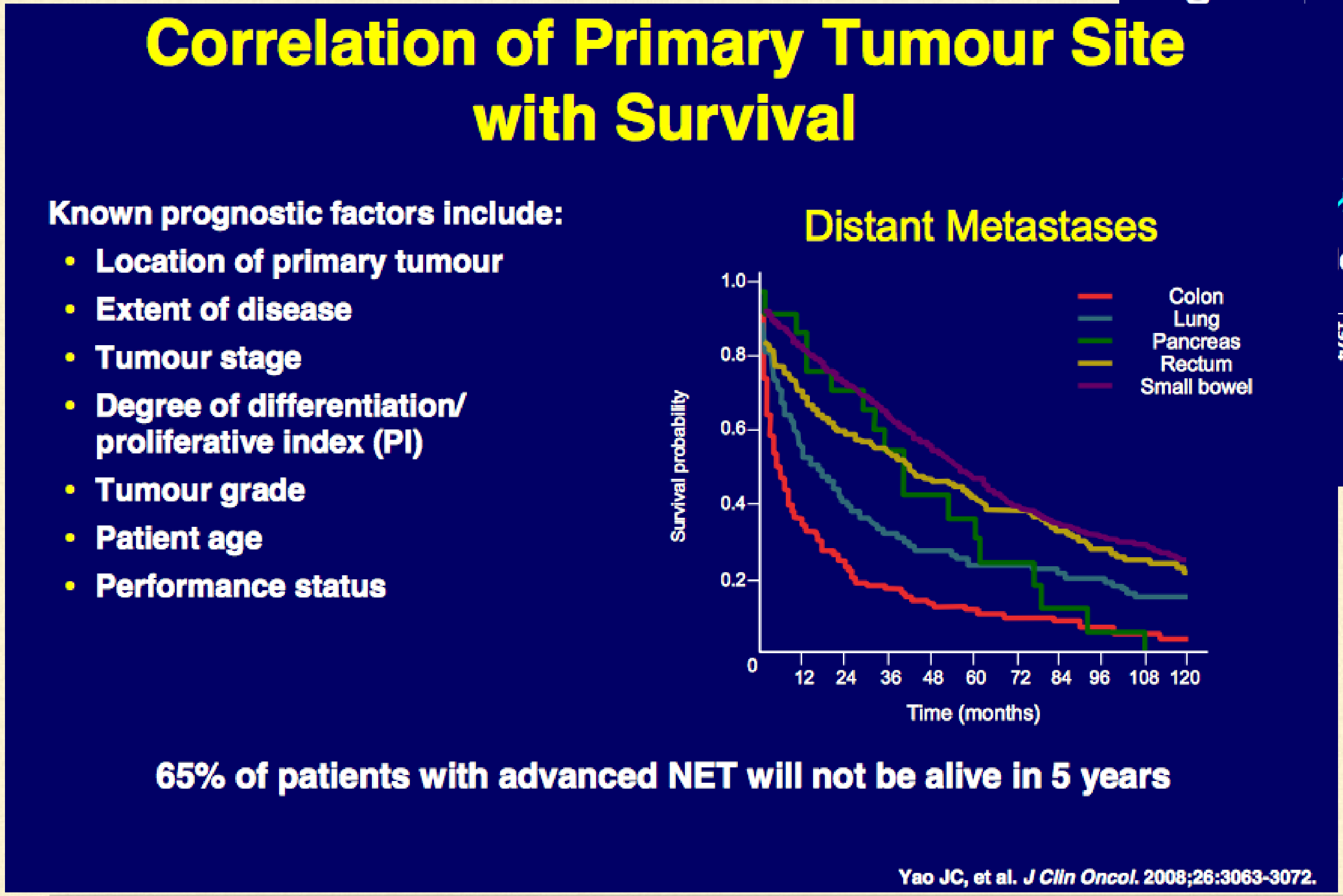

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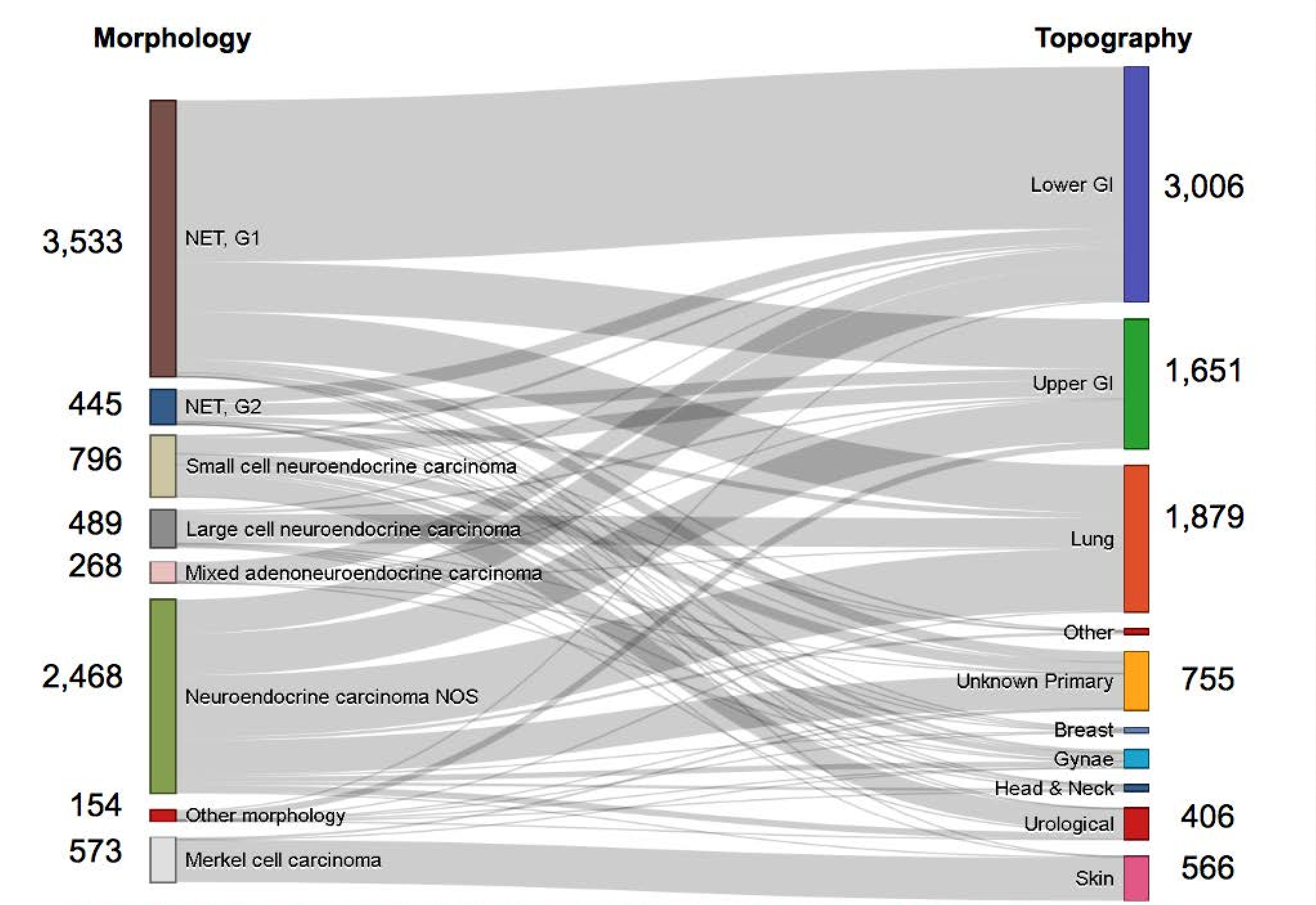

Figure 1: morphological and topological distribution of 8,726 neuroendocrine neoplasms diagnosed in England, 2013 and 2014

4,000 = 8 : 100,000 (England)

High % G1 NETs = Lower GI

High % G3 = Lung

G1 = <3% NET

G2 = 3-20% NET

G3 = >20% well / poorly

NET or NEC

nb > or <55%

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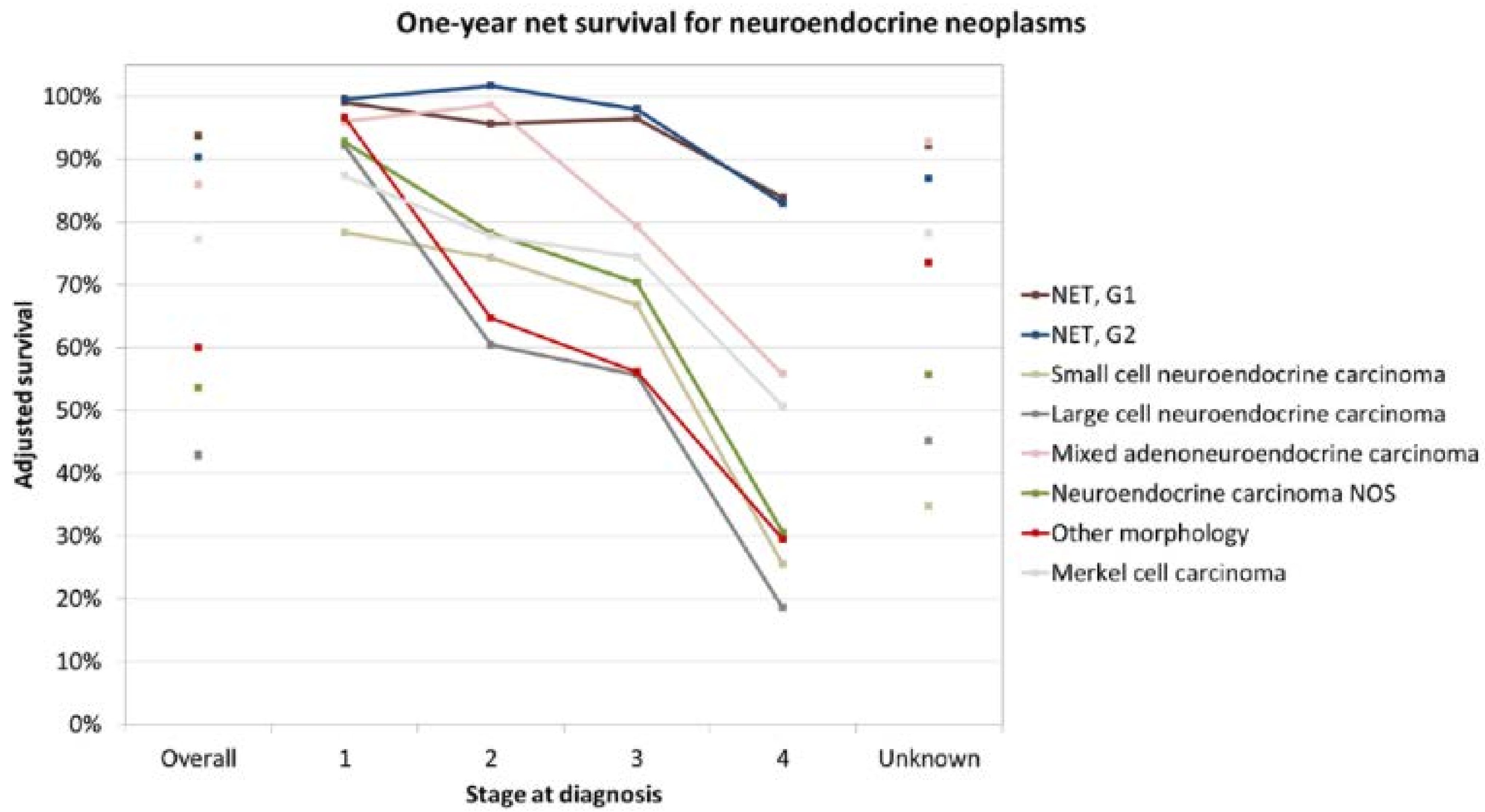

Figure 2: one year net survival for neuroendocrine neoplasms diagnosed in England, 2013-2014.

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Cancer – **excluding "less advanced" cancers**

Any malignant tumour positively diagnosed with histological confirmation and characterised by the uncontrolle

All cancers which are histologically classified as any of the following :

pre-malignant

non-invasive

cancer-in-situ

having either **borderline malignancy** or

having **low malignant potential**

All tumours of the prostate unless histologically classified as having a Gleason score of 7 or above or having p

Any skin cancer (including cutaneous lymphoma) other than malignant melanoma that has been histologically

http://www.aviva.co.uk/adviser/product-literature/files/da/da09003c.pdf : accessed February 2017

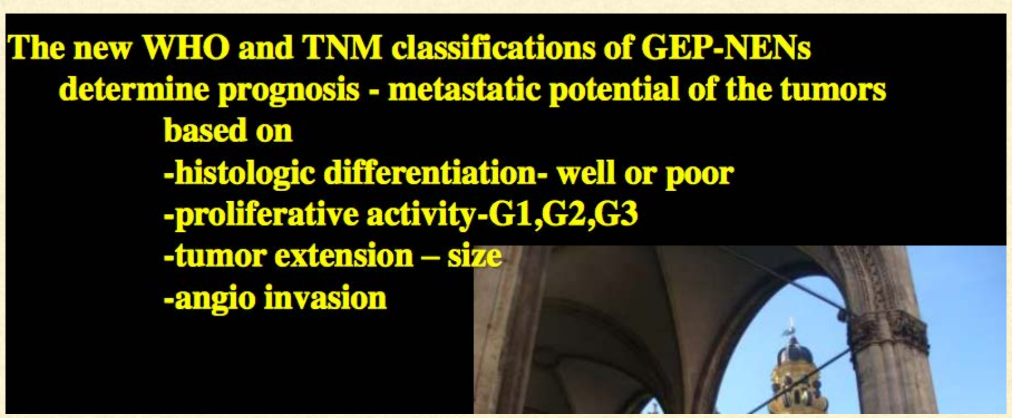
Malignant :
characterized by uncontrolled growth;
cancerous, invasive, or metastatic.
Immortal, invasive and itinerant !

NET : NEUROENDOCRINE TUMOURS

"Borderline malignancy" : potentially malignant cells that form in the tissue covering t

"Low malignant potential" : potentially malignant cells that form in the tissue covering

nb ENETs and AJCC :   T1 ≤1 cm;   invasion of muscularis propria



All NETs have malignant potential

"Terminal" : incurable, with a life expectancy / prognosis < 12 months.

http://www.aviva.co.uk/adviser/product-literature/files/da/da09003c.pdf : accessed February 2017

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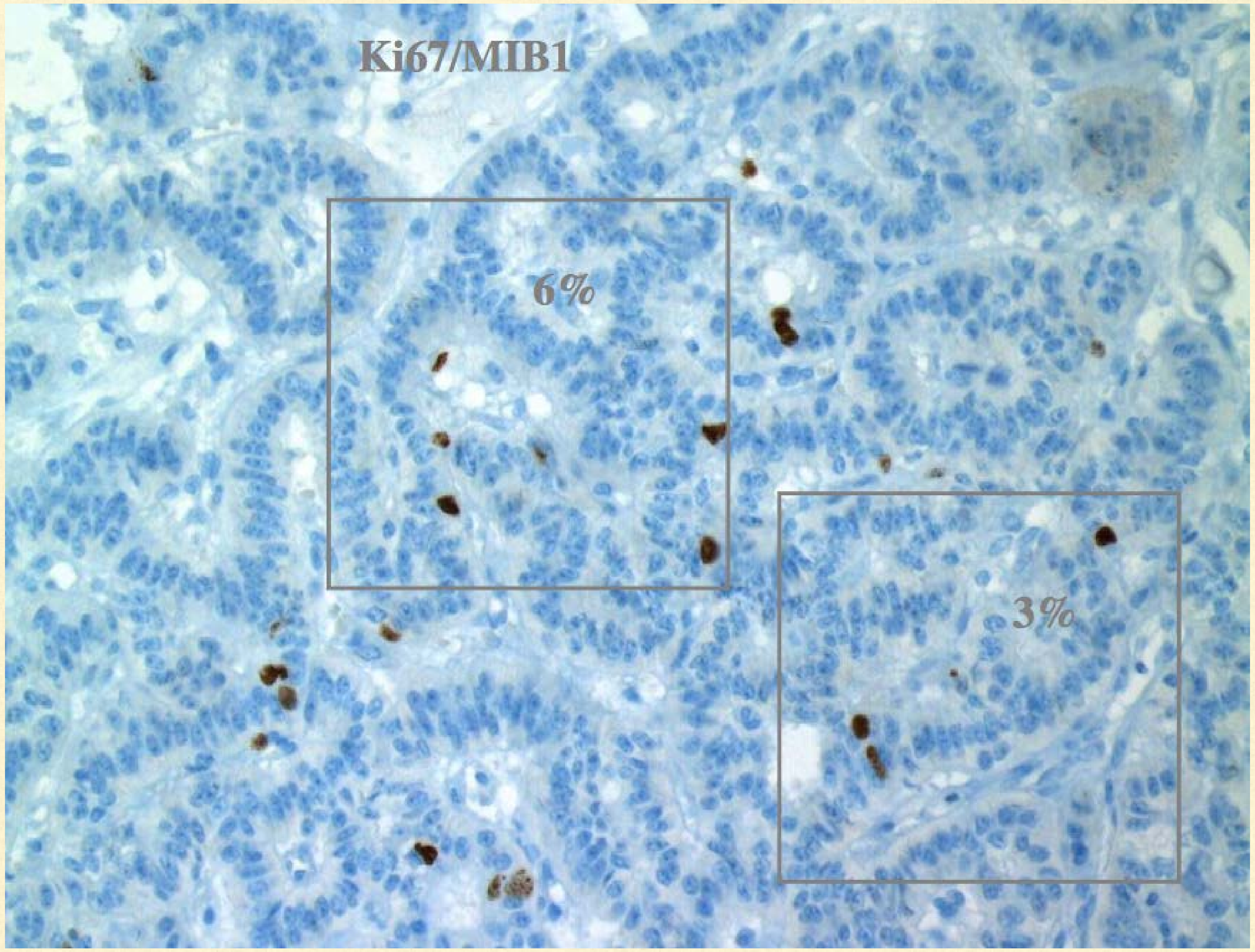

- Uniform, organised, nucleus to cell size ratio, communicate, react, specialise (mature), stick together, look alike, repair and/or self destruct (apoptosis) and die.
- Heterogenous, disorganised, large nucleus, ignore / interrupt cell signalling - don't communicate, no specialisation (immature), do not stick together, look different, faulty repair mechanism, over-ride self-destruct signals, immortal.
- NETs have the latter characteristics
- NETs can exhibit different behaviour between sites
- NETs can exhibit different behaviour within the same tumour

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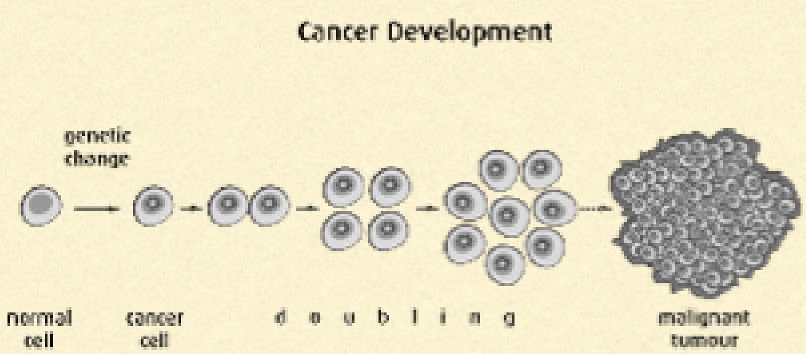
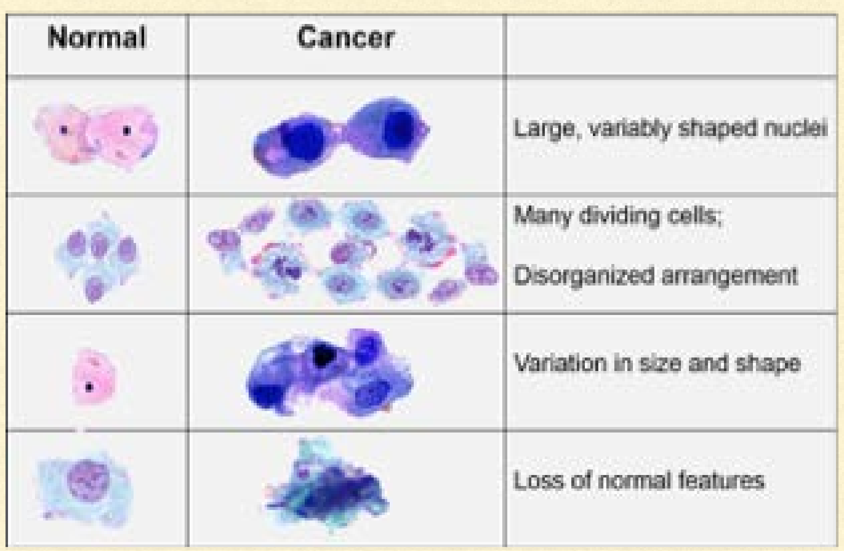
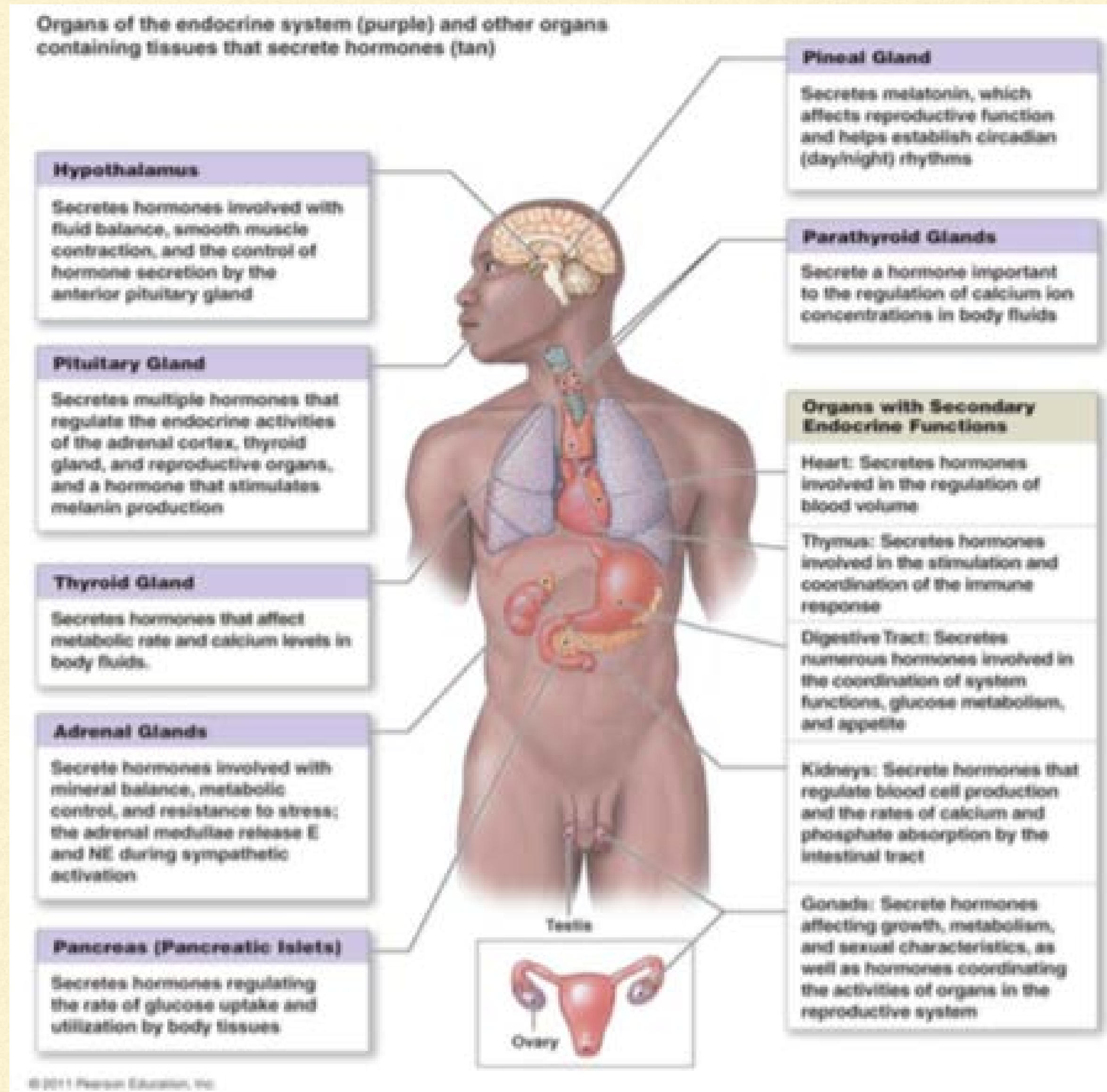
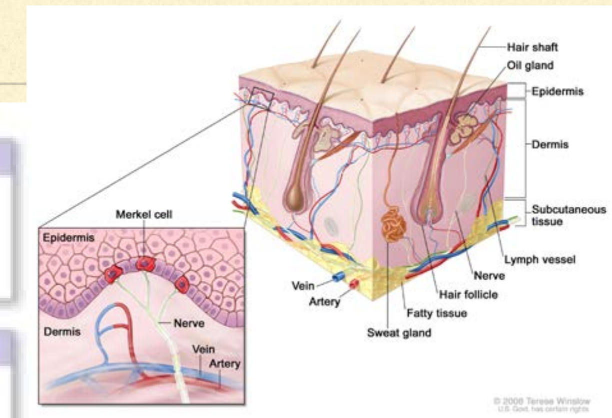
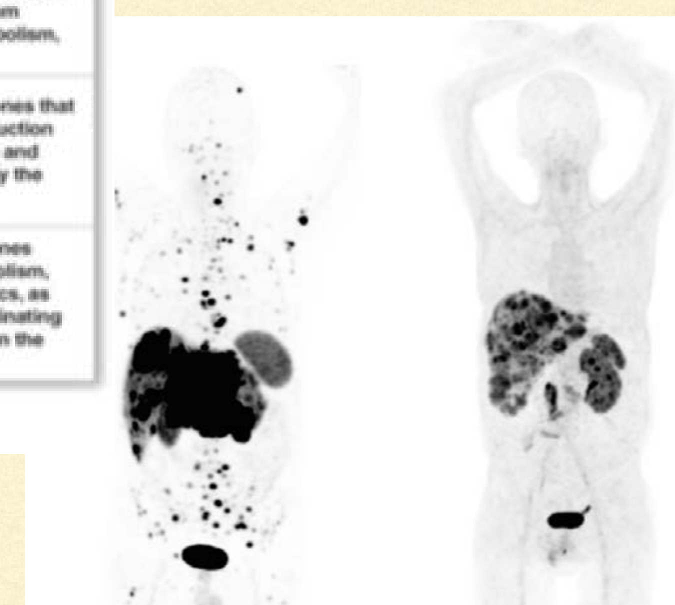


- NET is umbrella term









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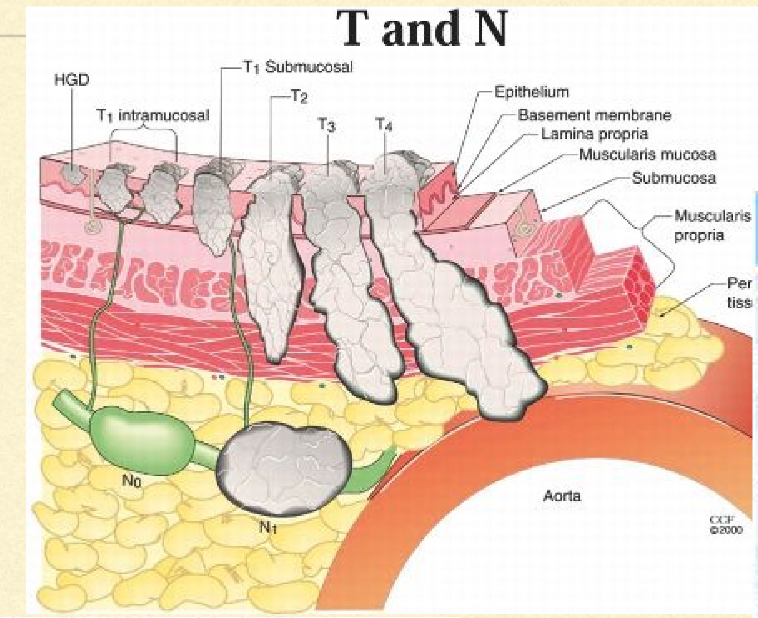
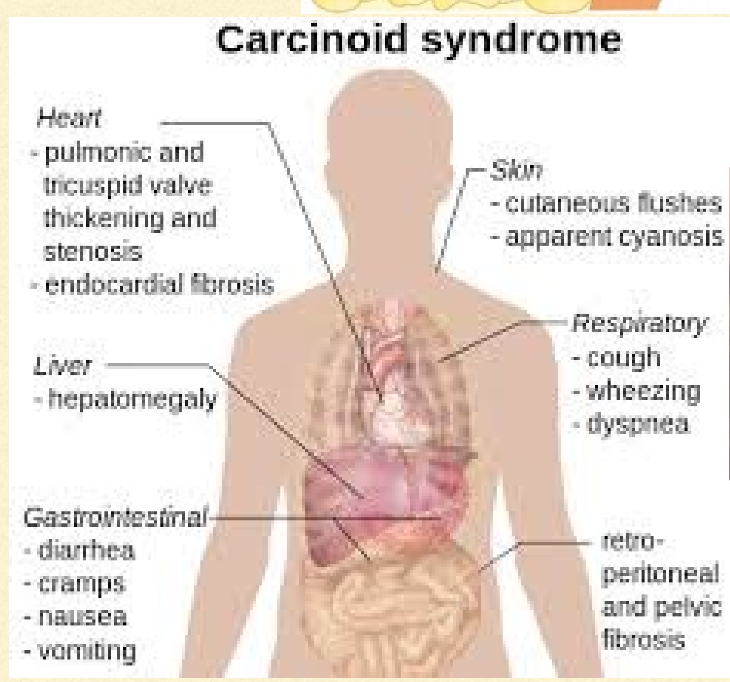
- Site
- Size
- TNM Staging
- Grading
- Function



	Well-differentiated		Poorly differentiated
Grade (ENETS)	Low (G1)	Intermediate (G2)	High (G3)
Ki-67 index (%)	≤2	3-20	>20
Anatomic imaging	more rapid growth on serial imaging		
Functional imaging	Octreoscan SPECT or SSTR PET +ve		FDG PET +ve
Prognosis	Indolent (slowly growing)		Aggressive
Treatment options	Surgery for localised +/- resectable metastatic disease		
	Observation Somatostatin analogues Radionuclide therapy		Chemotherapy
	Everolimus, sunitnib, α-interferon Liver metastases: radiofrequency ablation, hepatic embolisation, TACE, SIR-Spheres		



Differential Diagnosis (3)

Clinically PB can be distinguished from the following neuroendocrine tumors due to their different spectrum of symptoms:

Disease	Features
Insulinoma	Hypoglycemia, behavior change, weight gain and/or morning seizures
Gastrinoma	Severe gastrointestinal ulceration and diarrhea
VIPoma	Watery diarrhea, hyperkalemic and achlorhydia
Glucagonomas	Migratory necrolytic dermatitis, weight loss, stomatitis, anemia and hyperglycemia
Somatostatinomas	Diarrhea and may develop diabetes mellitus

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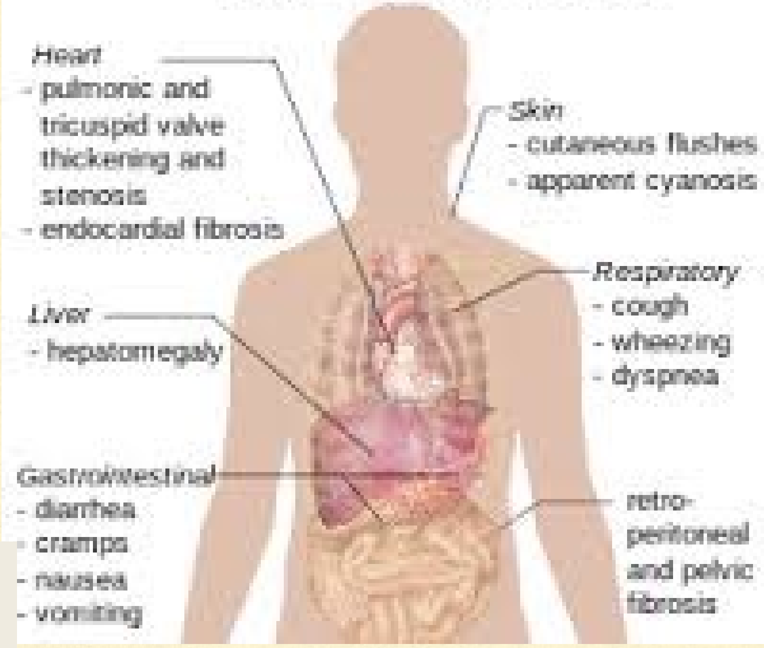
Carcinoid Heart Disease

- In patients with carcinoid tumors,
- Cardiac involvement, principally of the endocardium and valves of the right heart.
- The Carcinoid syndrome
  - distinctive episodic flushing of the skin and cramps,
  - nausea, vomiting, and diarrhea
  - bronchoconstrictive episodes resembling asthma.
  - **cardiac lesions** in about one-half (tricuspid and pulmonic valves may malfunction, with tricuspid regurgitation).

Carcinoid syndrome



Clinical Manifestations

- Carcinoid syndrome consists of a triad: flushing, diarrhoea and bronchospasm.
- Between 20-50% of all patients with carcinoid syndrome will develop carcinoid heart disease.
- Vasoactive substances such as 5-hydroxytryptamine produced by neoplastic cells are able to travel to the right heart via the hepatic vein/IVC and are thought to be responsible for deposition of endocardial plaques of fibrous tissue.
- Classically patients develop signs and symptoms of right heart failure: fatigue, oedema and ascites.

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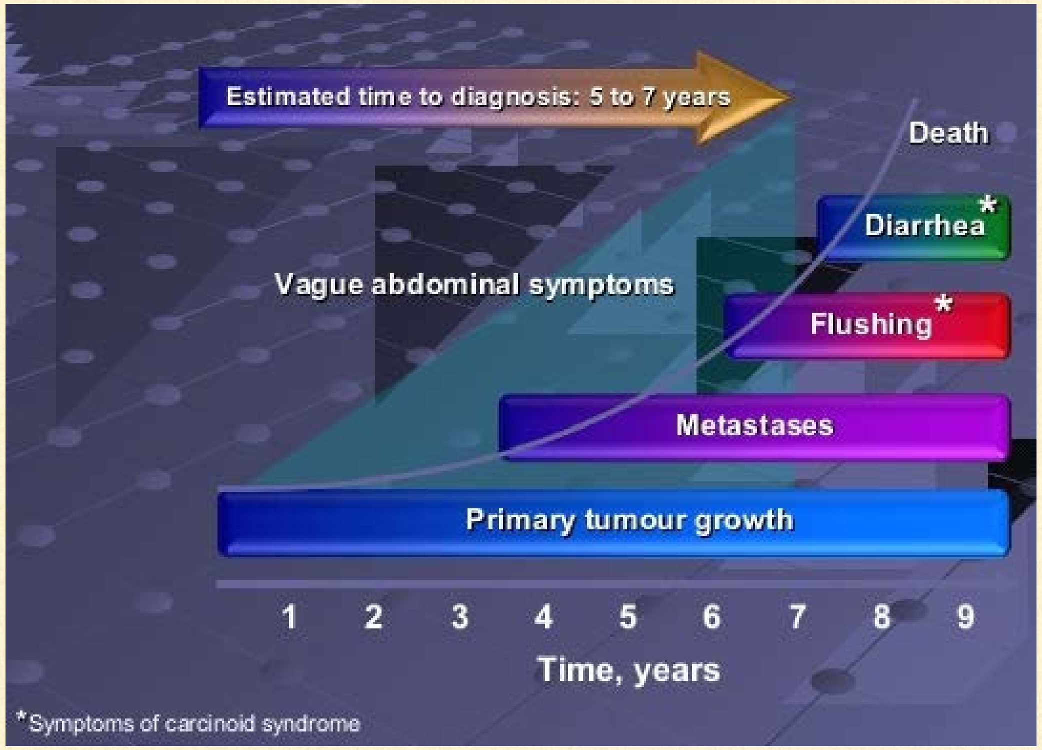

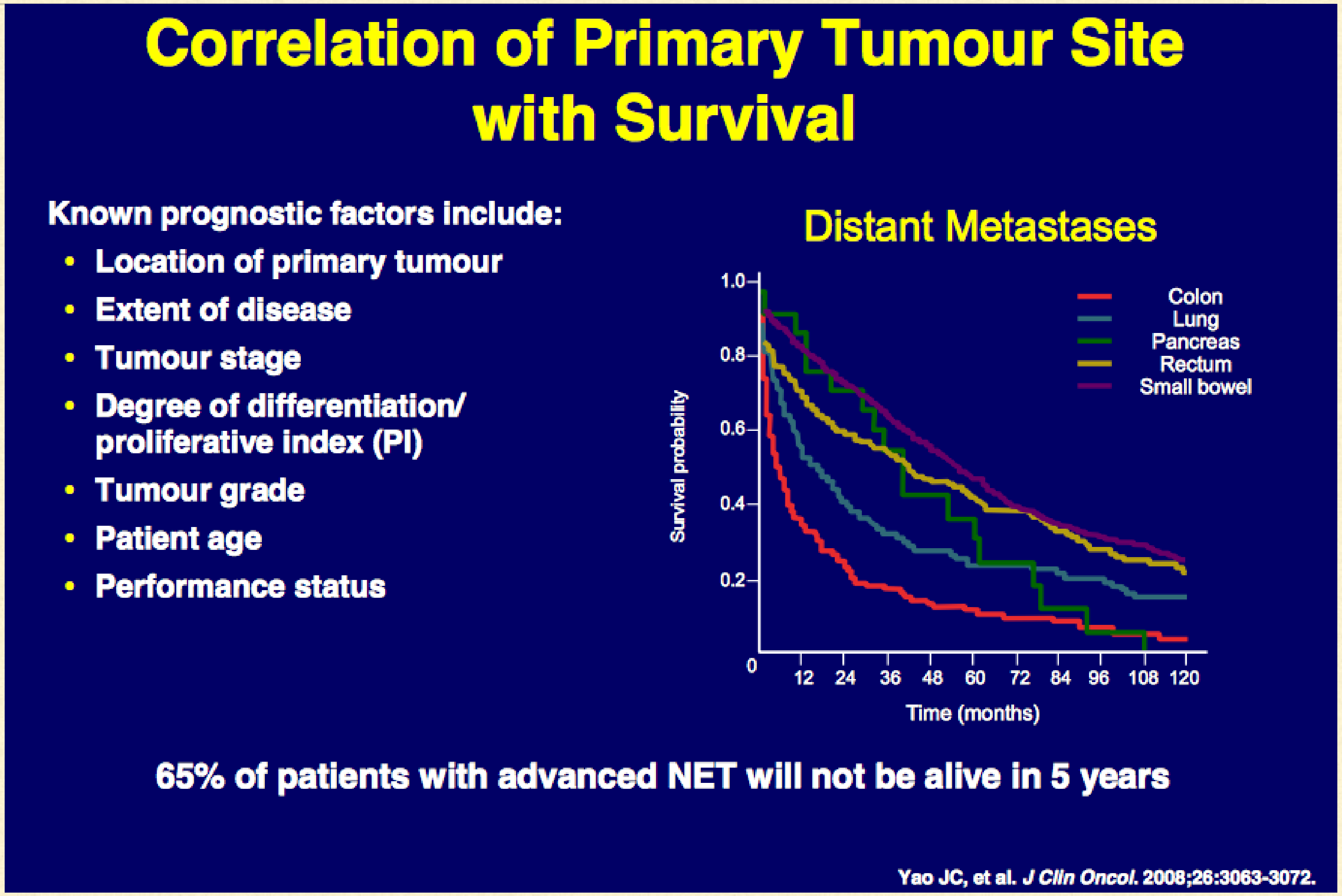

Disease

Treatment

Other Factors

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- www.netpatientfoundation.org
- www.ukinets.org/
- www.enets.org/